

which has simple poles at $z = \pm ci$, belongs to $\mathfrak{F}_a$ for all $0 < a < c$.

Another example is provided by $f(z) = 1/\cosh \pi z$, which belongs to $\mathfrak{F}_a$ whenever $|a| < 1/2$. This function, as well as one of its fundamental properties, was already discussed in Example 3, Section 2.1 of Chapter 3.

Note also that a simple application of the Cauchy integral formulas shows that if $f \in \mathfrak{F}_a$, then for every n , the n^{th} derivative of f belongs to $\mathfrak{F}_b$ for all b with $0 < b < a$ (Exercise 2).

Finally, we denote by $\mathfrak{F}$ the class of all functions that belong to $\mathfrak{F}_a$ for some a .

Remark. The condition of moderate decrease can be weakened somewhat by replacing the order of decrease of $A/(1+x^2)$ by $A/(1+|x|^{1+\epsilon})$ for any $\epsilon > 0$. As the reader will observe, many of the results below remain unchanged with this less restrictive condition.

2 Action of the Fourier transform on $\mathfrak{F}$

Here we prove three theorems, including the Fourier inversion and Poisson summation formulas, for functions in $\mathfrak{F}$. The idea behind all three proofs is the same: contour integration. Thus the approach used will be different from that of the corresponding results in Book I.

Theorem 2.1 *If f belongs to the class $\mathfrak{F}_a$ for some $a > 0$, then $|\hat{f}(\xi)| \leq Be^{-2\pi b|\xi|}$ for any $0 \leq b < a$.*

Proof. Recall that $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i x \xi} dx$. The case $b = 0$ simply says that $\hat{f}$ is bounded, which follows at once from the integral defining $\hat{f}$, the assumption that f is of moderate decrease, and the fact that the exponential is bounded by 1.

Now suppose $0 < b < a$ and assume first that $\xi > 0$. The main step consists of shifting the contour of integration, that is the real line, down by b . More precisely, consider the contour in Figure 1 as well as the function $g(z) = f(z)e^{-2\pi i z \xi}$.

We claim that as R tends to infinity, the integrals of g over the two vertical sides converge to zero. For example, the integral over the vertical segment on the left can be estimated by

$$\begin{aligned} \left| \int_{-R-ib}^{-R} g(z) dz \right| &\leq \int_0^b |f(-R-it)e^{-2\pi i(-R-it)\xi}| dt \\ &\leq \int_0^b \frac{A}{R^2} e^{-2\pi t \xi} dt \\ &= O(1/R^2). \end{aligned}$$

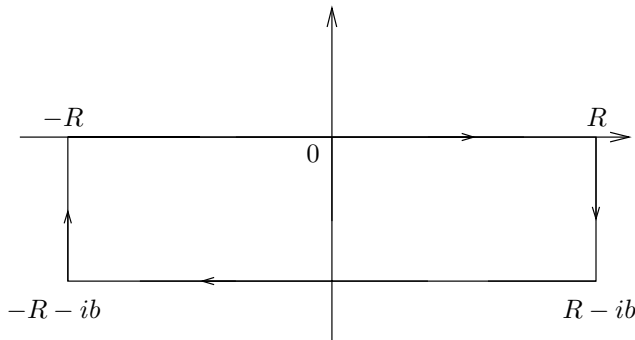


Figure 1. The contour in the proof of Theorem 2.1 when $\xi > 0$

A similar estimate for the other side proves our claim. Therefore, by Cauchy's theorem applied to the large rectangle, we find in the limit as R tends to infinity that

$$(1) \quad \hat{f}(\xi) = \int_{-\infty}^{\infty} f(x - ib)e^{-2\pi i(x - ib)\xi} dx,$$

which leads to the estimate

$$|\hat{f}(\xi)| \leq \int_{-\infty}^{\infty} \frac{A}{1 + x^2} e^{-2\pi b\xi} dx \leq B e^{-2\pi b\xi},$$

where B is a suitable constant. A similar argument for $\xi < 0$, but this time shifting the real line up by b , allows us to finish the proof of the theorem.

This result says that whenever $f \in \mathfrak{F}$, then $\hat{f}$ has rapid decay at infinity. We remark that the further we can extend f (that is, the larger a), then the larger we can choose b , hence the better the decay. We shall return to this circle of ideas in Section 3, where we describe those f for which $\hat{f}$ has the ultimate decay condition: compact support.

Since $\hat{f}$ decreases rapidly on $\mathbb{R}$, the integral in the Fourier inversion formula makes sense, and we now turn to the complex analytic proof of this identity.

Theorem 2.2 *If $f \in \mathfrak{F}$, then the Fourier inversion holds, namely*

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi \quad \text{for all } x \in \mathbb{R}.$$

Besides contour integration, the proof of the theorem requires a simple identity, which we isolate.

Lemma 2.3 *If A is positive and B is real, then $\int_0^\infty e^{-(A+iB)\xi} d\xi = \frac{1}{A+iB}$.*

Proof. Since $A > 0$ and $B \in \mathbb{R}$, we have $|e^{-(A+iB)\xi}| = e^{-A\xi}$, and the integral converges. By definition

$$\int_0^\infty e^{-(A+iB)\xi} d\xi = \lim_{R \rightarrow \infty} \int_0^R e^{-(A+iB)\xi} d\xi.$$

However,

$$\int_0^R e^{-(A+iB)\xi} d\xi = \left[-\frac{e^{-(A+iB)\xi}}{A+iB} \right]_0^R,$$

which tends to $1/(A+iB)$ as R tends to infinity.

We can now prove the inversion theorem. Once again, the sign of ξ matters, so we begin by writing

$$\int_{-\infty}^\infty \hat{f}(\xi) e^{2\pi i x \xi} d\xi = \int_{-\infty}^0 \hat{f}(\xi) e^{2\pi i x \xi} d\xi + \int_0^\infty \hat{f}(\xi) e^{2\pi i x \xi} d\xi.$$

For the second integral we argue as follows. Say $f \in \mathfrak{F}_a$ and choose $0 < b < a$. Arguing as the proof of Theorem 2.1, or simply using equation (1), we get

$$\hat{f}(\xi) = \int_{-\infty}^\infty f(u - ib) e^{-2\pi i (u-ib)\xi} du,$$

so that with an application of the lemma and the convergence of the integration in ξ , we find

$$\begin{aligned} \int_0^\infty \hat{f}(\xi) e^{2\pi i x \xi} d\xi &= \int_0^\infty \int_{-\infty}^\infty f(u - ib) e^{-2\pi i (u-ib)\xi} e^{2\pi i x \xi} du d\xi \\ &= \int_{-\infty}^\infty f(u - ib) \int_0^\infty e^{-2\pi i (u-ib-x)\xi} d\xi du \\ &= \int_{-\infty}^\infty f(u - ib) \frac{1}{2\pi b + 2\pi i(u-x)} du \\ &= \frac{1}{2\pi i} \int_{-\infty}^\infty \frac{f(u - ib)}{u - ib - x} du \\ &= \frac{1}{2\pi i} \int_{L_1} \frac{f(\zeta)}{\zeta - x} d\zeta, \end{aligned}$$

where L_1 denotes the line $\{u - ib : u \in \mathbb{R}\}$ traversed from left to right. (In other words, L_1 is the real line shifted down by b .) For the integral when $\xi < 0$, a similar calculation gives

$$\int_{-\infty}^0 \hat{f}(\xi) e^{2\pi i x \xi} d\xi = -\frac{1}{2\pi i} \int_{L_2} \frac{f(\zeta)}{\zeta - x} d\zeta,$$

where L_2 is the real line shifted up by b , with orientation from left to right. Now given $x \in \mathbb{R}$, consider the contour γ_R in Figure 2.

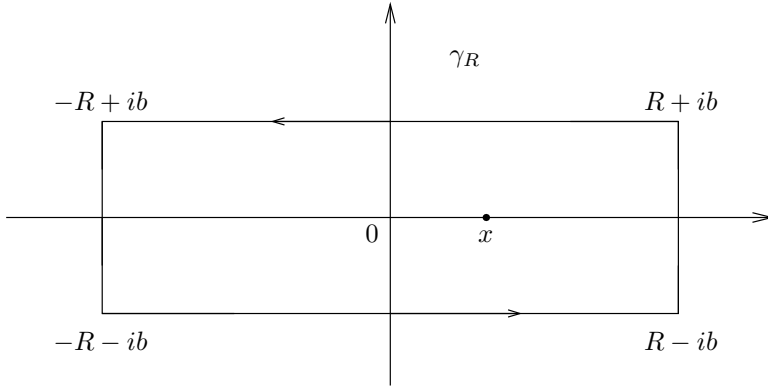


Figure 2. The contour γ_R in the proof of Theorem 2.2

The function $f(\zeta)/(\zeta - x)$ has a simple pole at x with residue $f(x)$, so the residue formula gives

$$f(x) = \frac{1}{2\pi i} \int_{\gamma_R} \frac{f(\zeta)}{\zeta - x} d\zeta.$$

Letting R tend to infinity, one checks easily that the integral over the vertical sides goes to 0 and therefore, combining with the previous results, we get

$$\begin{aligned} f(x) &= \frac{1}{2\pi i} \int_{L_1} \frac{f(\zeta)}{\zeta - x} d\zeta - \frac{1}{2\pi i} \int_{L_2} \frac{f(\zeta)}{\zeta - x} d\zeta \\ &= \int_0^\infty \hat{f}(\xi) e^{2\pi i x \xi} d\xi + \int_{-\infty}^0 \hat{f}(\xi) e^{2\pi i x \xi} d\xi \\ &= \int_{-\infty}^\infty \hat{f}(\xi) e^{2\pi i x \xi} d\xi, \end{aligned}$$

and the theorem is proved.

The last of our three theorems is the Poisson summation formula.

Theorem 2.4 *If $f \in \mathfrak{F}$, then*

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{n \in \mathbb{Z}} \hat{f}(n).$$

Proof. Say $f \in \mathfrak{F}_a$ and choose some b satisfying $0 < b < a$. The function $1/(e^{2\pi iz} - 1)$ has simple poles with residue $1/(2\pi i)$ at the integers. Thus $f(z)/(e^{2\pi iz} - 1)$ has simple poles at the integers n , with residues $f(n)/2\pi i$. We may therefore apply the residue formula to the contour γ_N in Figure 3 where N is an integer.

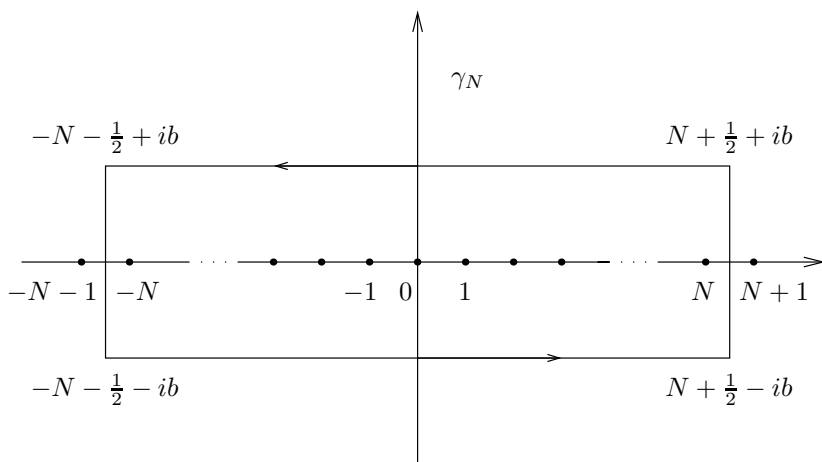


Figure 3. The contour γ_N in the proof of Theorem 2.4

This yields

$$\sum_{|n| \leq N} f(n) = \int_{\gamma_N} \frac{f(z)}{e^{2\pi iz} - 1} dz.$$

Letting N tend to infinity, and recalling that f has moderate decrease, we see that the sum converges to $\sum_{n \in \mathbb{Z}} f(n)$, and also that the integral over the vertical segments goes to 0. Therefore, in the limit we get

$$(2) \quad \sum_{n \in \mathbb{Z}} f(n) = \int_{L_1} \frac{f(z)}{e^{2\pi iz} - 1} dz - \int_{L_2} \frac{f(z)}{e^{2\pi iz} - 1} dz$$

where L_1 and L_2 are the real line shifted down and up by b , respectively.

Now we use the fact that if $|w| > 1$, then

$$\frac{1}{w-1} = w^{-1} \sum_{n=0}^{\infty} w^{-n}$$

to see that on L_1 (where $|e^{2\pi iz}| > 1$) we have

$$\frac{1}{e^{2\pi iz} - 1} = e^{-2\pi iz} \sum_{n=0}^{\infty} e^{-2\pi inz}.$$

Also if $|w| < 1$, then

$$\frac{1}{w-1} = - \sum_{n=0}^{\infty} w^n$$

so that on L_2

$$\frac{1}{e^{2\pi iz} - 1} = - \sum_{n=0}^{\infty} e^{2\pi inz}.$$

Substituting these observations in (2) we find that

$$\begin{aligned} \sum_{n \in \mathbb{Z}} f(n) &= \int_{L_1} f(z) \left(e^{-2\pi iz} \sum_{n=0}^{\infty} e^{-2\pi inz} \right) dz + \int_{L_2} f(z) \left(\sum_{n=0}^{\infty} e^{2\pi inz} \right) dz \\ &= \sum_{n=0}^{\infty} \int_{L_1} f(z) e^{-2\pi i(n+1)z} dz + \sum_{n=0}^{\infty} \int_{L_2} f(z) e^{2\pi inz} dz \\ &= \sum_{n=0}^{\infty} \int_{-\infty}^{\infty} f(x) e^{-2\pi i(n+1)x} dx + \sum_{n=0}^{\infty} \int_{-\infty}^{\infty} f(x) e^{2\pi inx} dx \\ &= \sum_{n=0}^{\infty} \hat{f}(n+1) + \sum_{n=0}^{\infty} \hat{f}(-n) \\ &= \sum_{n \in \mathbb{Z}} \hat{f}(n), \end{aligned}$$

where we have shifted L_1 and L_2 back to the real line according to equation (1) and its analogue for the shift down.

The Poisson summation formula has many far-reaching consequences, and we close this section by deriving several interesting identities that are of importance for later applications.

First, we recall the calculation in Example 1, Chapter 2, which showed that the function $e^{-\pi x^2}$ was its own Fourier transform:

$$\int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x \xi} dx = e^{-\pi \xi^2}.$$

For fixed values of $t > 0$ and $a \in \mathbb{R}$, the change of variables $x \mapsto t^{1/2}(x + a)$ in the above integral shows that the Fourier transform of the function $f(x) = e^{-\pi t(x+a)^2}$ is $\hat{f}(\xi) = t^{-1/2} e^{-\pi \xi^2/t} e^{2\pi i a \xi}$. Applying the Poisson summation formula to the pair f and $\hat{f}$ (which belong to $\mathfrak{F}$) provides the following relation:

$$(3) \quad \sum_{n=-\infty}^{\infty} e^{-\pi t(n+a)^2} = \sum_{n=-\infty}^{\infty} t^{-1/2} e^{-\pi n^2/t} e^{2\pi i n a}.$$

This identity has noteworthy consequences. For instance, the special case $a = 0$ is the transformation law for a version of the “theta function”: if we define ϑ for $t > 0$ by the series $\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$, then the relation (3) says precisely that

$$(4) \quad \vartheta(t) = t^{-1/2} \vartheta(1/t) \quad \text{for } t > 0.$$

This equation will be used in Chapter 6 to derive the key functional equation of the Riemann zeta function, and this leads to its analytic continuation. Also, the general case $a \in \mathbb{R}$ will be used in Chapter 10 to establish a corresponding law for the more general Jacobi theta function Θ .

For another application of the Poisson summation formula we recall that we proved in Example 3, Chapter 3, that the function $1/\cosh \pi x$ was also its own Fourier transform:

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{\cosh \pi x} dx = \frac{1}{\cosh \pi \xi}.$$

This implies that if $t > 0$ and $a \in \mathbb{R}$, then the Fourier transform of the function $f(x) = e^{-2\pi i a x} / \cosh(\pi x/t)$ is $\hat{f}(\xi) = t / \cosh(\pi(\xi + a)t)$, and the Poisson summation formula yields

$$(5) \quad \sum_{n=-\infty}^{\infty} \frac{e^{-2\pi i a n}}{\cosh(\pi n/t)} = \sum_{n=-\infty}^{\infty} \frac{t}{\cosh(\pi(n+a)t)}.$$

This formula will be used in Chapter 10 in the context of the two-squares theorem.

3 Paley-Wiener theorem

In this section we change our point of view somewhat: we do not suppose any analyticity of f , but we do assume the validity of the Fourier inversion formula

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi \quad \text{if} \quad \hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx,$$

under the conditions $|f(x)| \leq A/(1+x^2)$ and $|\hat{f}(\xi)| \leq A'/(1+\xi^2)$. For a proof of the inversion formula under these conditions, we refer the reader to Chapter 5 in Book I.

We start by pointing out a partial converse to Theorem 2.1.

Theorem 3.1 *Suppose $\hat{f}$ satisfies the decay condition $|\hat{f}(\xi)| \leq A e^{-2\pi a|\xi|}$ for some constants $a, A > 0$. Then $f(x)$ is the restriction to $\mathbb{R}$ of a function $f(z)$ holomorphic in the strip $S_b = \{z \in \mathbb{C} : |\operatorname{Im}(z)| < b\}$, for any $0 < b < a$.*

Proof. Define

$$f_n(z) = \int_{-n}^n \hat{f}(\xi) e^{2\pi i \xi z} d\xi,$$

and note that f_n is entire by Theorem 5.4 in Chapter 2. Observe also that $f(z)$ may be defined for all z in the strip S_b by

$$f(z) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi z} d\xi,$$

because the integral converges absolutely by our assumption on $\hat{f}$: it is majorized by

$$A \int_{-\infty}^{\infty} e^{-2\pi a|\xi|} e^{2\pi b|\xi|} d\xi,$$

which is finite if $b < a$. Moreover, for $z \in S_b$

$$\begin{aligned} |f(z) - f_n(z)| &\leq A \int_{|\xi| \geq n} e^{-2\pi a|\xi|} e^{2\pi b|\xi|} d\xi \\ &\rightarrow 0 \quad \text{as } n \rightarrow \infty, \end{aligned}$$

and thus the sequence $\{f_n\}$ converges to f uniformly in S_b , which, by Theorem 5.2 in Chapter 2, proves the theorem.

We digress briefly to make the following observation.